PROJECT SELECTION

Healthcare and Biotech Proposals, Research and Implementation

Based on the PubMed and OpenAlex starter kit | English edition

Recommended: ResearchBridge

Start with a healthcare or biotech question, retrieve traceable papers, explain methods and limitations, and explore institution links supported by publications. A first version focused on digital health and care transitions offers a coherent demonstration.

This document presents three primary proposals and two extension ideas, including users, product scope, implementation steps, technology choices and evaluation. The Chinese and English editions use the same structure.

Proposal	Use case	Data readiness
ResearchBridge	Evidence and research collaboration discovery	Closest fit; relationship enrichment required
CareBridge	Evidence and checklist support for care transitions	Relevant seeds; operational workflows needed
TrialBridge	Multilingual clinical research navigation	Trial registry data required
Wearable Evidence Lab	Wearable evidence and validation planning	Targeted literature expansion required
Coverage Guide	MassHealth renewal information and tasks	Official rules and administrative information required

Reading guide: data limits on page 2; proposals on pages 3-6; engineering, evaluation and delivery on pages 7-9; references on page 10.

01 / DATA AUDIT

What can the two datasets support?

All nine source files and every row of both CSVs were reviewed. The six supplied screenshots provide topic context.

Item	PubMed [D1]	OpenAlex [D2]
Unit and size	One bibliographic record per row; 60 rows, 5 columns	One institution record per row; 135 rows, 4 columns
Available fields	pmid, title, journal, published, authors	institution, city, works, cited_by
Basic quality	No blank fields; no duplicate PMIDs or exact duplicate rows	No blank fields or exact duplicate rows
Key gaps	No abstracts, full text, affiliations, DOIs or trial records	No papers, topics, authors, years, OpenAlex IDs or ROR IDs

Four issues to resolve before building

- Date semantics: published strings begin with 2024 in 32 records, 2025 in 26 and 2026 in 2. The README describes 2024. Reconcile print, electronic and indexing dates rather than assuming the records are wrong. [R2]

- No direct join: the CSVs share no key. Author-name strings cannot establish affiliations. Institution names need stable IDs and hierarchy checks.

- Fetch scripts are not reconstruction pipelines: PubMed requests 20 IDs; OpenAlex retrieves only the first 25 institutions without pagination. Both save fresh.json rather than rebuilding the complete CSVs.

- Interpretation: the institution list mixes types and organizational levels. Counts have no annual breakdown here. They do not establish care quality, current specialization, growth or a deduplicated statewide publication total.

The READMEs state a save date of 2026-10-18; this is a documented date, not an independently verified retrieval timestamp.

02 / PRIMARY PROPOSAL ONE

ResearchBridge

Research evidence and collaboration navigation | Recommended first MVP

Users and problem

Graduate researchers, biotech R&D teams and healthcare innovators need to understand available methods, evidence limitations and relevant institutions. The product connects these questions through a traceable evidence path.

Product and demonstration

A user asks how digital tools could support care transitions. Results include papers, research cards, method comparisons and verified affiliation links. Seed titles include a medication-literacy chatbot (PMID 41726533) and caregiver support (41726516).

Implementation steps

1. Preserve the raw CSV, deduplicate by PMID and use EFetch to add abstracts, DOIs, publication types and available affiliations. [R1]

2. Resolve an OpenAlex work by PMID or DOI, read institution IDs from authorships and reconcile them with an enriched institution directory. [R3-R4]

3. Establish keyword retrieval first, then add embeddings. Generate methods, limitations and item-level citations only from retrieved evidence.

4. Show unmatched and missing-data states. Review ambiguous institutions manually. Title-only records support bibliographic information and provisional categorization, not study-result summaries.

Technology	Purpose
Python / pandas / SQLite	Cleaning, deduplication, provenance and relational storage
Keyword search + embeddings	Compare semantic retrieval against a keyword baseline
LLM API + Streamlit	Cited research cards and an interactive search prototype

Evaluate top-five relevance, citation support, paper-institution linkage accuracy and time to find usable evidence. These are proposed measurements, not demonstrated outcomes.

03 / PRIMARY PROPOSAL TWO

CareBridge

Evidence and checklist support for care transitions

Users and problem

Serve care coordinators, social workers and community health workers. Focus on a single transition, such as returning to the community from a nursing facility, and organize questions about handoffs, follow-up, medication understanding and service connections.

Data seeds and product

PMID 41726541 concerns a mobile-app prototype for community transition after lower-limb fracture; 41726533 concerns medication literacy; 41001595 concerns postpartum insurance changes. These are research leads, not proof that a proposed workflow or intervention is effective. [D1]

Demonstration: use a synthetic case

A coordinator enters a fictional discharge scenario. The system organizes information to confirm, relevant studies and a draft workflow. Each item identifies whether it comes from research, an official process or a pending human check.

Implementation and technology

- Interview practitioners and map the current process and common omissions. Do not let the model invent a standard operating procedure.

- Store research, organization-approved workflows and service directories separately, with scope, versions and update timestamps.

- Use Python, SQLite, structured checklist templates and retrieval-grounded generation. A rules layer manages required fields and task states; the model explains text.

- Keep the first version to drafts and confirmation workflows. Real referrals, scheduling or transport coordination require operational data and institutional integrations.

Evaluate required-item coverage on synthetic cases, erroneous or unsupported items, task time and clarity. Qualified professionals retain clinical decisions.

04 / PRIMARY PROPOSAL THREE

TrialBridge

Multilingual clinical research navigation

Users and value

Help patients, families and referring staff find studies worth asking about and understand participation requirements, locations and unresolved conditions. Support discovery and questions without issuing final eligibility decisions.

Why additional data is essential

The starter includes a title about accessibility of skull-base-disorder trials (PMID 41140426). A paper does not establish current recruitment, and an institution city is not a trial location. ClinicalTrials.gov registry data is required. [R6]

Additional fields	Purpose
NCT ID / study title	Stable study identity and a link to the official record
Recruitment status / update date	Show registered recruitment status and freshness
Eligibility criteria	Explain criteria and questions that require confirmation
Locations / contacts	Support navigation using actual study sites and contact routes

Implementation and technology

- Retrieve registry records through the API, cache responses and preserve update dates. Separate institution addresses from study sites.

- Use structured filters for status and location. Preserve unknown states for missing or ambiguous criteria rather than guessing.

- Generate multilingual explanations and questions for the study team while preserving original criteria. Start in Streamlit; separate the frontend and API when needed.

Evaluate faithful explanations, stale status handling, location accuracy, multilingual back-checking and abstention on missing data. Study teams verify final eligibility.

05 / EXTENSION IDEAS

Wearable health and coverage support

Potential follow-on products; the starter CSVs do not yet supply their essential data.

Wearable Evidence Lab

Positioning: help R&D teams ask how sensing or wearable technologies have been validated, in which populations and with which measurements. Biosensing (PMID 41669754) and bioelectronics (41058940) titles are seeds, not individual wearable signals.

- Implementation: expand a targeted literature search. Extract modality, population, reference standard, validation method and limitations from available abstracts or permitted full text, then have researchers review them.

- Technology: retrieval, structured extraction, schema validation and comparison tables. Risk prediction would additionally require time series, device metadata, outcome labels and participant-level validation.

Research limit: absence from a small literature set does not establish a research gap. A reproducible search and screening strategy is needed.

Coverage Guide

Positioning: turn official MassHealth notices and renewal steps into understandable explanations, document checklists and questions to ask. The insurance-change paper (PMID 41001595) is a topic lead, not an authority on program rules.

- Implementation: collect current official rules, notice templates, language versions and assistance routes. Add document-text extraction and human confirmation if individual notices are later supported.

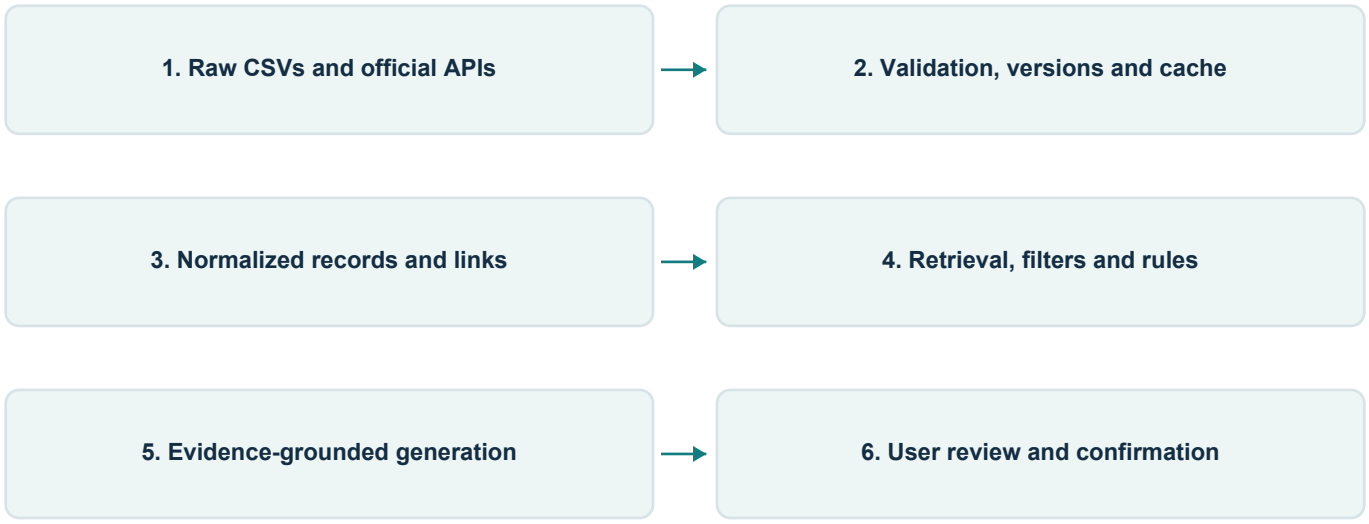
- Technology: document parsing, version-aware retrieval, rules-based tasks and multilingual explanations. Extract case deadlines only from notices or official sources; mark unsupported deadlines for confirmation.

This document does not validate current renewal frequency, deadlines or eligibility rules. Verify official policy before pursuing this proposal; the state retains eligibility decisions.

06 / TECHNICAL ARCHITECTURE

Build the evidence layer first

A small prototype does not require training a large model or starting with a separate vector database.



Run stages 1 through 6. Stage 3 links records through enriched PMID/DOI and institution IDs; it is not a direct merge of the two starter CSVs.

Table	Key fields and purpose
papers	pmid, doi, title, abstract, publication_dates, source_url
institutions	institution_id, ror, name, city, parent_ids, source_url
paper_institutions	pmid, institution_id, mapping source and review status
evidence_cards	paper_id, extracted fields, supporting passages, model and version
source_snapshots	source, query parameters, retrieval time and raw-response reference

Technology choices: Python for data work, SQLite for prototype storage, keyword retrieval as a baseline, embeddings for semantic similarity, an LLM for grounded synthesis and Streamlit for the interface. Consider FastAPI and PostgreSQL once scale and multi-user needs are established.

Engineering requirements: preserve unknowns, deduplicate papers by stable ID, prevent multi-institution double counting, and use caching, pagination and retries. Check current API authentication and limits when implementing. [R5]

07 / RESEARCH AND EVALUATION

How will we know it helps?

The following is a proposed study design, not a completed product experiment.

Research process

1. Define the problem: select one user group and task. Interview users about existing research practices, time costs and what makes an output usable.

2. Assemble evidence: record search queries, dates and inclusion/exclusion criteria. Enrich abstracts and affiliations while retaining unmatched and disputed records.

3. Label a benchmark: create representative questions and relevance judgments. Have two reviewers independently assess a subset and reconcile disagreements.

4. Compare methods: evaluate keyword, vector and hybrid retrieval on the same candidate corpus using the same questions and scoring rules.

5. Test with users: compare existing search with the prototype on matched tasks. Observe time, errors, completion and confusion. A small pilot identifies usability issues; it does not establish clinical benefit.

Measure	Definition or assessment
Precision@5	Human-judged relevant results among the first five / 5, for benchmark tasks with five returned results
Citation support	Verifiable claims supported by their citations / reviewed claims
Linkage accuracy	Correct paper-institution links / reviewed links; report unmatched coverage separately
Task performance	Completion, time, errors and omissions, supplemented by interviews

Test missing abstracts, unresolved and same-name institutions, multi-institution papers, conflicting sources, stale trials and translations that alter criteria. No performance target or outcome has been achieved or reported.

08 / BUILD AND DELIVERY

Complete one verifiable user journey

The schedule is an illustrative plan for a small team, subject to staffing, API access and data quality.

Stage	Work	Reviewable deliverable
Days 1-2	Scope the task; audit data; enrich abstracts and dates	Data dictionary, provenance and missing-data list
Days 3-4	Build bibliographic search and research cards	Keyword baseline and links to source records
Days 5-6	Enrich institution IDs and paper links	Link results, unmatched states and review notes
Days 7-8	Compare retrieval and add grounded answers	Benchmark results and failure cases
Days 9-10	Run a user pilot, refine the interface and demo	Pilot notes, known limits and next-version backlog

If only 48 hours are available

Limit ResearchBridge to question entry, paper retrieval, source cards and a small set of manually verified institution links. If abstracts or mappings are unavailable, disclose the gap and stop at bibliographic search instead of inventing conclusions.

Scope and completion criteria

- Deliver a runnable prototype, data dictionary and provenance, repeatable retrieval/cleaning scripts, evaluation results and demonstration cases.

- Completion requires traceable institution recommendations, supported study claims, explicit unknowns and resolved or disclosed review disagreements.

- Expand only after establishing ResearchBridge usefulness, then choose trials, care workflows or wearable evidence as the next focused extension.

09 / SOURCES AND SCOPE

Data and implementation references

Prepared on 2026-10-24. Official links support this planning research; recheck current documentation before implementation.

Local data and context

[D1] pubmed/data.csv, README.md, fetch.py and fetch.js: 60 bibliographic records and source-fetch behavior.
[D2] openalex/data.csv, README.md, fetch.py and fetch.js: 135 institution records and source-fetch behavior.
[D3] Root README.md and the six user-supplied Healthcare & biotech topic screenshots.

[R1] **NCBI E-utilities**
ESearch retrieves identifiers; EFetch retrieves full records. Fields may remain missing.
https://www.ncbi.nlm.nih.gov/books/NBK25497/

[R2] **PubMed Help**
Publication-date fields and print/electronic date interpretation.
https://pubmed.ncbi.nlm.nih.gov/help/#pubdate

[R3] **OpenAlex: Get Singleton**
Resolve a work using PMID, DOI or OpenAlex ID.
https://help.openalex.org/api/get-single-entities/

[R4] **OpenAlex: Authorships**
Authors, raw affiliations and institution links.
https://help.openalex.org/data/authorships/

[R5] **OpenAlex: Authentication**
Authentication, budgets and paging requirements; consult current rules.
https://help.openalex.org/api/authentication/

[R6] **ClinicalTrials.gov API record example**
Example of the official registry structure; not a recommended study.
https://clinicaltrials.gov/api/v2/studies/NCT02993146

Interpretation and use

Proposals, architecture, schedule and measures are suggested designs. No prototype, user study or clinical-outcome evaluation has been completed. Research claims require supporting evidence. Qualified professionals retain clinical decisions, study teams confirm trial eligibility and the state determines MassHealth eligibility. Check source permissions before publicly redistributing abstracts or full text.